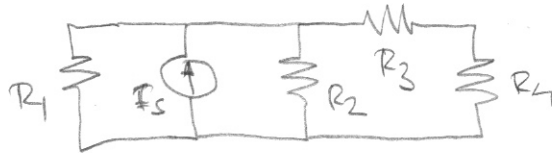


3.67)

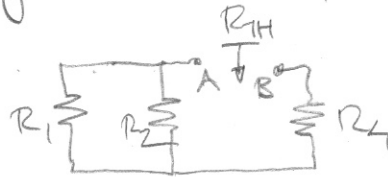


$$\begin{aligned} R_1 &= 10 \Omega \\ R_2 &= 9 \Omega \\ R_3 &= 4 \Omega \\ R_4 &= 4 \Omega \\ I_S &= 2 \text{ A} \end{aligned}$$

BESTÄM EN NORTONEKVIVALENT FÖR R_3 (LAST).

BESTÄM ÄVEN EFFEKTEN SOM DISSIPERAS AV R_3 .

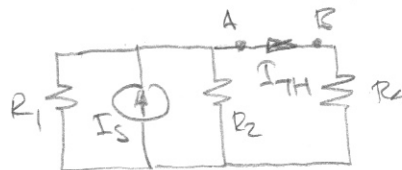
Lösning: FÖRST R_{TH}



$$R_{TH} = R_4 + \frac{R_1 R_2}{R_1 + R_2} =$$

$$= 4 + \frac{10 \cdot 9}{10 + 9} = \underline{8.74 \Omega}$$

NÄRREFTER I_N



SPÄNNINGEN I DEN
ÖVRE NODEN BESTÄMMER
STRÖMMEN GENOM
 R_4 .

SPÄNNINGEN BLIR

$$V_A = \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4} \right]^{-1} I_S = \left[\frac{1}{10} + \frac{1}{9} + \frac{1}{4} \right]^{-1} 2 =$$

$$= \underline{4.34 \text{ V}}$$

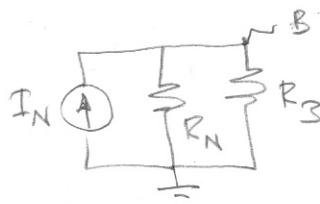
och strömmen I_N

$$I_N = \frac{V_A}{R_4} = \frac{4.34}{4} = \underline{1.08 \text{ A}}$$

$$\underline{\text{SVAR:}} \quad \begin{cases} R_{TH} = R_N = 8.74 \Omega \\ I_N = 1.08 \text{ A} \end{cases}$$

ÄSSÄ
RESUR

HUR STOR EFFEKT ÅTER R_3 JPP?



$$V_B = \frac{R_N R_3}{R_N + R_3} I_N = 2.96 \text{ V}$$

EFFEKT

$$P = VI = \frac{V^2}{R} \Rightarrow P_{R_3} = \frac{V_B^2}{R_3} = \frac{2.96^2}{4} \approx$$

$$\underline{\underline{2.19 \text{ WATT}}}$$